CSE563 — Unit II Numerical Worksheet

Vectors and Matrices

40 questions • CO2 • Student Worksheet

Instructions

Attempt all 40 questions and show your working. Where a check is available — trace and determinant for eigenvalues, multiplying an inverse back to I, confirming interpolation weights sum to 1 — *perform it*; it costs one line and catches most arithmetic slips. Refer to the *Unit II Numerical Guide* for a worked example of each problem type.

A. Vectors (Q1–Q7)

Q1. For $\mathbf{u} = (3, -1, 2)$ and $\mathbf{v} = (1, 4, -2)$, compute $\mathbf{u} + \mathbf{v}$, $2\mathbf{u} - 3\mathbf{v}$ and $\mathbf{u} \cdot \mathbf{v}$.

Q2. Find $\|\mathbf{a}\|$ and the unit vector along $\mathbf{a} = (2, -3, 6)$.

Q3. Find the angle between $\mathbf{u} = (1, 1, 0)$ and $\mathbf{v} = (1, 0, 1)$.

Q4. Determine whether $\mathbf{a} = (2, -1, 3)$ and $\mathbf{b} = (1, 5, 1)$ are orthogonal. If not, find the projection of $\mathbf{a}$ onto $\mathbf{b}$.

Q5. Find the projection of $\mathbf{u} = (4, 3)$ onto $\mathbf{v} = (1, 0)$, and interpret the result geometrically.

Q6. Determine whether $(1, 2, 1)$, $(2, 1, 0)$, $(1, -1, -1)$ are linearly independent.

Q7. Compute $\mathbf{u} \times \mathbf{v}$ for $\mathbf{u} = (1, 0, 2)$, $\mathbf{v} = (0, 3, 1)$, and verify the result is orthogonal to both.

B. Matrices (Q8–Q17)

Q8. For $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$, compute AB and BA.

Q9. Compute A^2 for $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, and state $\det(A^2)$ without further multiplication.

Q10. Evaluate $\det \begin{pmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \\ -1 & 4 & 1 \end{pmatrix}$.

Q11. Find A^{-1} for $A = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}$ and verify $AA^{-1} = I$.

Q12. Determine whether $A = \begin{pmatrix} 3 & 6 \\ 2 & 4 \end{pmatrix}$ is invertible. Justify.

Q13. Solve $\begin{cases} 3x + 2y = 16 \\ x - y = 2 \end{cases}$ using the matrix inverse.

Q14. Solve $\begin{cases} x + 2y = 8 \\ 3x - y = 3 \end{cases}$ using Cramer's rule.

Q15. Find the eigenvalues of $A = \begin{pmatrix} 6 & -2 \\ -2 & 9 \end{pmatrix}$ and verify them using the trace and determinant.

Q16. Find the eigenvalues and corresponding eigenvectors of $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$, and confirm the eigenvectors are orthogonal.

Q17. Find the eigenvalues of the triangular matrix $A = \begin{pmatrix} 2 & 5 & 1 \\ 0 & 3 & 4 \\ 0 & 0 & 7 \end{pmatrix}$.

Q18. For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, verify that $(A^T)^{-1} = (A^{-1})^T$.

C. Principal Component Analysis (Q18–Q23)

Q19. The data are $(2, 1), (4, 3), (6, 5)$. Compute the mean vector and the centred data matrix.

Q20. For the centred data of the previous question, compute the 2×2 covariance matrix and its eigenvalues. Comment on what the result says about the data.

Q21. Compute the covariance matrix of $(1, 2), (2, 4), (3, 6), (4, 8)$ and find both eigenvalues.

Q22. Given $S = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$, find both eigenvalues, the first principal component (normalised), and the percentage of variance it explains.

Q23. Using $\mathbf{w}_1 = (0.8944, 0.4472)$ from the previous question, project the centred point $(3, -1)$ onto PC1 and onto PC2 $= (-0.4472, 0.8944)$. Verify that the squared projections sum to the squared length of the point.

Q24. A PCA on a 5-feature dataset yields eigenvalues $12.0, 5.5, 1.8, 0.5, 0.2$. How many components are needed to retain at least 90% of the variance?

D. Linear Discriminant Analysis (Q24–Q27)

Q25. $C_1 = \{(2,3), (3,4), (4,5)\}$ and $C_2 = \{(6,2), (7,3), (8,4)\}$. Compute both class means and the between-class mean difference $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$.

Q26. For the data of the previous question, compute S_1, S_2 and S_W.

Q27. C_1 has $\boldsymbol{\mu}_1 = (1,2)$, C_2 has $\boldsymbol{\mu}_2 = (5,6)$, and $S_W = \begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix}$. Find the LDA direction $\mathbf{w}$ and explain why it is not simply parallel to $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$.

Q28. Using $\mathbf{w} = (4, 1)$, $\boldsymbol{\mu}_1 = (1, 2)$ and $\boldsymbol{\mu}_2 = (5, 6)$ from the previous question, compute the decision threshold and classify the new point $(3, 4)$.

E. Image Transformation (Q28–Q35)

Q29. Write the 3×3 homogeneous matrix that translates by $(5, -2)$, and apply it to $(3, 7)$.

Q30. Write the matrix that scales by $(3, 0.5)$ about the origin and apply it to the square with vertices $(0, 0), (2, 0), (2, 2), (0, 2)$.

Q31. Rotate the point $(4, 0)$ by $60°$ anticlockwise about the origin. Give the exact and decimal answer.

Q32. A point is first rotated by $90°$ anticlockwise about the origin, then translated by $(2, 3)$. Find the single composite matrix and apply it to $(1, 0)$.

Q33. Repeat the previous question with the operations in the opposite order (translate by $(2, 3)$ first, then rotate by $90°$). Compare the two results.

Q34. Find the single matrix that rotates by 90° anticlockwise about the point $(2, 1)$, and apply it to $(4, 1)$.

Q35. Write the matrix for a shear of the x-coordinate by factor $k = 2$, and find the image of the unit square $(0, 0), (1, 0), (1, 1), (0, 1)$. What is the area of the image?

Q36. A transformation is given by $M = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix}$. Find M^{-1} and use it to recover the original point whose image is $(9, 5)$.

F. Image Interpolation (Q36–Q40)

Q37. Pixel values are $I(0, 0) = 50$, $I(1, 0) = 100$, $I(0, 1) = 150$, $I(1, 1) = 200$. Estimate $I(0.5, 0.5)$ by bilinear interpolation.

Q38. For the same four pixels, estimate $I(0.8, 0.3)$ by bilinear interpolation. Show the four weights and verify they sum to 1.

Q39. For the pixels of Q36, give the nearest-neighbour estimate at $(0.8, 0.3)$ and state the difference from the bilinear answer.

Q40. The 1-D row $[10, 50, 30, 70]$ is enlarged to 7 samples. Give the nearest-neighbour and linear-interpolated outputs.

Q41. A 4×4 image is to be rotated by $30°$. Explain with a calculation why interpolation is unavoidable, using the destination pixel $(3, 1)$ and inverse mapping about the origin.